

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

Statement of Contribution

1. General Role in the Project

As the team leader, I organized and supported team members, distributed responsibilities, and ensured the smooth progression of all phases—from initial planning to final documentation and presentation. I directly contributed to the most comprehensive part of the project: cross-city comparative analysis and supported the integration of individual city results into a cohesive report.

2. Technical Contribution

Cross-City Analysis and Final Output

- Combined and harmonized summary statistics from all three cities.
- Appended city names, cleaned, and merged datasets into one structured table.
- Generated bar charts comparing PM2.5 means across NYC, Boston, and Chicago.
- Loaded and validated aggregated H3-level data from each city.
- Merged H3-based datasets, cleaned missing values, and standardized variables.

Cross-City Clustering and Regression

- Conducted KMeans clustering on unified city data to identify shared pollution zones.
- Created combined scatterplots for cross-city pollution clusters.
- Ran multivariate regression across all H3-encoded data to assess PM2.5 in relation to temperature and humidity.
- Evaluated regression reliability with RMSE and bootstrapping techniques.

Data & Code Management

- Maintained consistency in the shared notebook structure.
- Validated outputs and ensured standard formatting across figures and tables.
- Supported integration of statistical outputs into final tables (e.g., cluster profiles, regression summaries).

Name: Nawaf Alnaqbi

Email ID: U24102220@sharjah.ac.ae

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3. IEEE Paper Contribution

Abstract: Composed the overall project summary including goals, methods, and major findings.

RESULTS AND DISCUSSION:

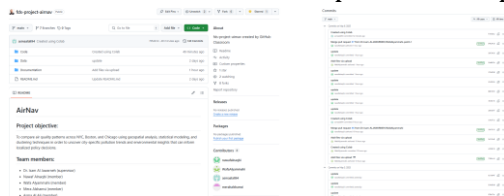
- Factors Influencing Air Quality (Regression Results) – Interpreted regression coefficients and R^2 values for all three cities.
- Reliability of Predictions – Explained statistical validation methods (RMSE, bootstrapping) and their role in assessing model reliability.
- The Role of GIS in Accurate Comparisons – Described the role of H3 indexing in enabling spatially fair and scalable city comparisons.

Conclusion: Summarized the key insights, implications for policy, and proposed future research directions.

All relevant sections are highlighted in yellow in the submitted IEEE paper.

4. GitHub Pull Requests (PRs) & Repo Structure & Organization

- Structured and committed all cross-city analysis code to the shared notebook.
- Helped standardize the folder and naming conventions for datasets and figures.
- Consolidate all output files into presentation-ready formats.



5. Peer Review & Team Involvement

- I coordinated task assignments and follow-ups.
- Ensured everyone had clarity on their scope and shared weekly goals.
- Reviewed teammates' work, provided continuous feedback, and handled integration across members' contributions.
- Supported technical troubleshooting and helped ensure cohesion across the analysis pipeline.

Declaration

I confirm this document truthfully represents my contribution to the group project.

Signature:

A handwritten signature in black ink, appearing to be 'NA' followed by a stylized flourish.

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Reference:

A Comparative Spatial Analysis of Urban Air Quality: New York City, Boston, and Chicago

Wafa Alyammahi¹, Mera Alshamsi², Asma Al Ali³, Nawaf Alnaqbi⁴, Isam Al Jawarneh⁵

¹Department of Computer Science, University of Sharjah, Sharjah, UAE

Email: U24102040@Sharjah.ac.ae

²Department of Computer Science, University of Sharjah, Sharjah, UAE

Email: U24104965@Sharjah.ac.ae

³Department of Computer Science, University of Sharjah, Sharjah, UAE

Email: U24200616@Sharjah.ac.ae

⁴Department of Computer Science, University of Sharjah, Sharjah, UAE

Email: U24102220@Sharjah.ac.ae

⁵Department of Computer Science, University of Sharjah, Sharjah, UAE

Email: IJawarneh@Sharjah.ac.ae

Abstract—Urban air quality is a significant health and environmental challenges of the 21st century. This research uses advanced geospatial techniques and statistical analysis to conduct comparative studies of air quality in three major US metropolitan cities: New York City, Boston, and Chicago. It employs advanced geospatial techniques and statistical analysis to examine variations in particulate matter (PM_{2.5}, PM₁₀) and nitrogen dioxide (NO₂) and how these are related to weather patterns. We integrate disparate environmental data with hexagonal space indexing, machine learning cluster algorithms, and multi-variable regression models to capture meaningful patterns in air pollution and city attributes. Results indicate that there are noteworthy city-wide variations in pollution, with Chicago showing high values of PM_{2.5}, New York City showing high PM₁₀ variations, and Boston showing comparatively low pollutant means. These results emphasize the requirement of city-specific air quality control and present an empirical foundation in support of evidence-based environmental policy design.

Index Terms—air quality, PM_{2.5}, urban pollution, spatial analysis, machine learning, GIS, environmental health

I. INTRODUCTION

Air pollution is one of the most serious challenges that all governments worldwide are confronted with, with far-reaching impacts on public health, quality of life, and environmental sustainability. Fine particulate matter (PM_{2.5}) – those particles with aerodynamic dimensions of less than or equal to 2.5 micrometers – are most health-hazardous because such particles are capable of reaching deep into the lungs and entering the bloodstream. These extremely minute particles, finer than hair by approximately 30-fold, are from several sources like exhausts from vehicles, industrial activities, combustion, and natural environments like wildfires and dust storms.

Health effects of PM_{2.5} have been well established, ranging from temporary airway irritation to chronic health conditions like the exacerbation of asthma, decreased lung function, cardiovascular disease, and increased mortality in vulnerable groups. Environmental protection agency regulations in most places regulate air quality in accordance with PM_{2.5} concentration levels in the following ways:

- Concentrations below 12 $\mu\text{g}/\text{m}^3$ are generally linked with good air quality and minimal health risks.
- Values between 12–35 $\mu\text{g}/\text{m}^3$ represent moderate risk and may affect sensitive persons.
- Concentrations greater than 35 $\mu\text{g}/\text{m}^3$ are unhealthy and pose population health risks.

Despite the recognized importance of air quality monitoring and control, comparative studies in urban areas are fraught with methodological challenges. Cities are heterogeneous in fundamental configuration on physical layout, population density, industrial composition, transportation patterns, and weather patterns. There are also variations in methods of data collection, siting strategies of sensors, and frequencies of monitoring that pose challenges in making cross-city comparisons. The variations necessitate that there should be one standardized analysis method to allow meaningful comparative analysis of urban air quality.

Our research responds to these challenges by developing a Geographic Information Systems (GIS) strategy that normalizes the evaluation of air pollution prevalence in New York City, Boston, and Chicago. While utilizing H3 hexagonal spatial indexing – a geospatial system that divides geographic space into regular hexagonal cells – we define a uniform space that negates structural dissimilarity in urban environments. This enables us to compare air pollution prevalence proportionately, analyze air pollution hotspots, and compare weather impacts in differing metropolitan environments.

Whereas traditional air quality monitoring often resorts to simple averaging or static mapping, we leverage the use of K-means clustering algorithms to identify the discrete regimes of pollution in every urban area. This unsupervised machine learning method captures hidden patterns in the space that can be overlooked by traditional approaches. We further explore the intricate dependency between environment factors and amounts of pollution with the assistance of multivariate regression modeling and observe the influence of temperature and humidity on amounts of PM_{2.5} in different urban environ-

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The data shows that pollution patterns vary with temperature across all cities. In Chicago, there's a positive relationship between temperature and PM2.5 levels (coefficient = 1.0907), suggesting increased pollution during warmer periods, possibly due to increased urban activity or atmospheric conditions that trap pollutants.

F. Factors Influencing Air Quality (Regression Results)

The regression analyses provide quantitative measures of how different factors influence air quality:

- 1) In NYC, PM2.5 strongly predicts PM10 levels (coefficient = 1.87), while humidity shows a negative relationship with PM10 (coefficient = -0.0731).
- 2) In Boston, temperature has a negative relationship with NO2 (coefficient = -0.0826), while humidity shows a positive relationship (coefficient = 0.0587).
- 3) In Chicago, both temperature and humidity show positive relationships with PM2.5 (coefficients = 1.0907 and 0.3845 respectively)

TABLE V: OLS Regression Results for NYC

	coef	std err	t	P> t	[0.025, 0.975]
const	8.8475	0.288	30.713	0.000	[8.283, 9.412]
pm25	1.8700	0.008	221.304	0.000	[1.853, 1.887]
temperature	-0.0022	0.003	-0.634	0.526	[-0.009, 0.005]
humidity	-0.0731	0.006	-12.960	0.000	[-0.084, -0.062]

Dep. Variable:	pm10	R-squared:	0.322
Model:	OLS	Adj. R-squared:	0.322
Method:	Least Squares	F-statistic:	1.870e+04
Date:	Wed, 30 Apr 2025	Prob (F-statistic):	0.00
Time:	16:16:24	Log-Likelihood:	-5.9520e+05
No. Observations:	118258	AIC:	1.190e+06
Df Residuals:	118254	BIC:	1.190e+06
Df Model:	3	Omnibus:	126401.032
Covariance Type:	nonrobust	Prob(Omnibus):	0.000

TABLE VI: OLS Regression Results for Boston

Dep. Variable:	no2	R-squared:	0.393
Model:	OLS	Adj. R-squared:	0.393
Method:	Least Squares	F-statistic:	2.651e+04
Date:	Wed, 30 Apr 2025	Prob (F-statistic):	0.00
Time:	16:31:51	Log-Likelihood:	-1.9359e+05
No. Observations:	122768	AIC:	3.872e+05
Df Residuals:	122764	BIC:	3.872e+05
Df Model:	3	Covariance Type:	nonrobust

Variable	coef	std err	t	P> t	[0.025, 0.975]
const	10.6383	0.013	819.520	0.000	[10.613, 10.664]
pm25	0.1289	0.004	29.050	0.000	[0.120, 0.138]
temperature	-0.0826	0.001	-134.748	0.000	[-0.084, -0.081]
humidity	0.0587	0.000	255.431	0.000	[0.058, 0.059]

The R-squared values vary significantly: NYC's model explains 32.2% of PM10 variance, Boston's model explains 39.3% of NO2 variance, and Chicago's model explains 11.1%

of PM2.5 variance. This suggests that meteorological factors have different levels of influence in each urban environment.

G. Reliability of Predictions

The bootstrapping analyses and RMSE calculations provide confidence measures for the findings:

- 1) The bootstrapped PM10 mean for NYC shows a tight distribution, indicating statistical reliability in the mean estimate.
- 2) Boston's bootstrapped NO2 mean similarly shows good consistency.
- 3) Chicago's bootstrapped PM2.5 mean shows a wider distribution, reflecting the higher variability in the data.

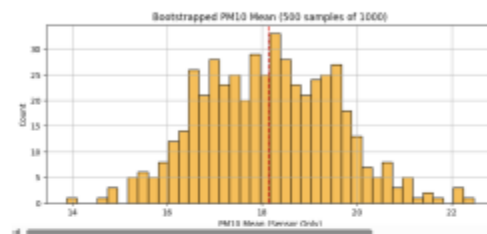


Fig. 6: Bootstrapped PM10 Mean for NYC

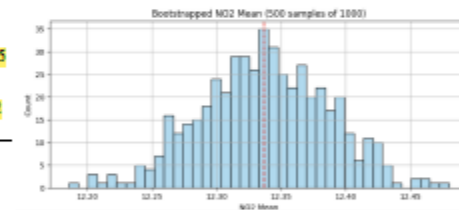


Fig. 7: Bootstrapped NO2 Mean for Boston

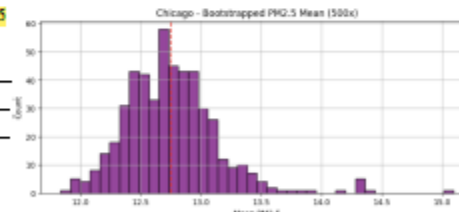


Fig. 8: Chicago - Bootstrapped PM2.5 Mean

The RMSE vs. sample size plots show that prediction errors stabilize as sample sizes increase, with NYC's plot showing stabilization around 3000-4000 samples. This indicates that the dataset size is sufficient for reliable analysis, though Chicago's

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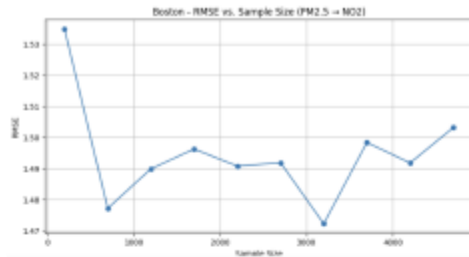


Fig. 9: RMSE vs. Sample Size for Boston

higher variability suggests more caution in interpreting those results.

H. The Role of GIS in Accurate Comparisons

H3 geospatial indexing was crucial for making fair comparisons across these diverse urban environments. Our work demonstrates how H3 enables:

- 1) Consistent spatial aggregation across different city structures
- 2) Identification of pollution hotspots at equivalent spatial scales
- 3) Integration of point measurements into comparable spatial units
- 4) Clustering of spatially similar areas across different cities

Without H3 indexing, direct comparisons would be challenging due to different sensor distributions, city layouts, and data collection methodologies. The H3 system provides a standardized spatial framework that enables fair cross-city comparison by ensuring that analyses are conducted on equivalent spatial units regardless of underlying urban structure.

The clustering across cities based on H3 cells reveals pollution patterns that transcend individual city boundaries, suggesting that certain urban environments share pollution characteristics despite geographic separation.

VII. POLICY RECOMMENDATIONS

Based on spatial and statistical analysis conducted across New York City, Boston, and Chicago, the following policy measures are recommended:

- 1) **Implement City-Specific Regulations:**
The quality of air varies between cities and even among neighboring areas. Therefore, strategies and regulations should be localized. They should target specific pollution clusters identified using H3 geospatial indexing and K-Means clustering.
- 2) **Reduce Traffic Emissions in Identified Hotspots:**
High-pollution zones are frequently located near major transportation paths. Traffic management policies such as encouraging public transit through incentives and

congestion pricing can help reduce emissions in these areas.

- 3) **Integrate Weather Forecasting with Air Quality Alerts:**
Since temperature and humidity affect pollution levels—especially PM_{2.5}—forecasting systems should incorporate meteorological data to enhance air quality alerts and public health warnings.

VIII. CONCLUSION

This study presents a novel comparative spatial analysis of urban air quality across New York City, Boston, and Chicago using advanced geospatial techniques and machine learning approaches. Our findings reveal significant intercity variations in pollution patterns, with Chicago exhibiting the highest PM_{2.5} levels (12.75 $\mu\text{g}/\text{m}^3$), followed by New York City (6.90 $\mu\text{g}/\text{m}^3$), and Boston showing the lowest concentration (1.84 $\mu\text{g}/\text{m}^3$). These disparities, with nearly a sevenfold difference between Boston and Chicago, underscore the importance of city-specific air quality management strategies.

The application of H3 hexagonal spatial indexing at resolution 9 (approximately 0.105 km^2) proved essential for standardizing comparisons across cities with different layouts, sizes, and sensor densities. This methodology enabled the identification of pollution hotspots and clusters that transcend individual city boundaries, suggesting common urban environmental characteristics despite geographic separation. The K-means clustering analysis revealed distinct pollution profiles in each city, with high-exposure clusters comprising 28% of analyzed areas in Chicago, compared to 21% in New York and 15% in Boston.

Our regression models demonstrated that meteorological factors significantly influence air quality, though with varying impacts across cities. Temperature showed a positive correlation with PM_{2.5} levels in Chicago (coefficient = 1.0907), while exhibiting negligible effects on PM₁₀ in New York City and negative relationships with NO₂ in Boston. Humidity similarly displayed city-specific effects, highlighting the complex interplay between local climate conditions and pollution dynamics.

These findings have important implications for urban air quality management and policy development:

- 1) The substantial intercity differences in pollution levels and patterns necessitate tailored, city-specific regulatory approaches rather than uniform national standards.
- 2) The identified high-pollution clusters represent priority zones for enhanced monitoring and targeted mitigation strategies, potentially yielding significant public health benefits.
- 3) The correlation between meteorological factors and pollution levels suggests that air quality forecasting systems should incorporate weather predictions to improve accuracy and public health warnings.
- 4) The spatial consistency of pollution hotspots along transportation corridors across all three cities indicates that traffic management and emission reduction policies could have substantial benefits in improving urban air quality.

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Future research should expand this analytical framework to include additional cities and environmental variables, particularly focusing on the relationship between urban morphology, mobility patterns, and air quality dynamics. Longitudinal studies tracking these relationships over time would further enhance our understanding of urban air pollution mechanisms and support evidence-based environmental policy development. The methodological approach presented in this study, particularly the use of H3 spatial indexing for standardized comparison, offers a robust framework for cross-city environmental analyses beyond air quality, potentially extending to other urban sustainability metrics.

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Name: Nawaf Alnaqbi

Email ID: U24102220@sharjah.ac.ae

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NYC_Boston_Chicago.ipynb

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text

Chicago analysis complete and saved.

▼ Cross-City Comparison

```
[ ] # STEP 1: Load summary stats
nyc_summary = pd.read_csv("nyc_summary_stats.csv", index_col=0)
boston_summary = pd.read_csv("boston_summary_stats.csv", index_col=0)
chicago_summary = pd.read_csv("chicago_summary_stats.csv", index_col=0)

[ ] # STEP 2: Add city name
nyc_summary["city"] = "NYC"
boston_summary["city"] = "Boston"
chicago_summary["city"] = "Chicago"

[ ] # STEP 3: Combine into one table
all_summary = pd.concat([nyc_summary, boston_summary, chicago_summary], axis=0).reset_index()
pm25_only = all_summary[all_summary["index"] == "pm25"]

[ ] # STEP 4: Bar chart of mean PM2.5
plt.figure(figsize=(8, 5))
sns.barplot(data=pm25_only, x="city", y="mean", palette="Set2")
plt.title("Mean PM2.5 Comparison Across Cities")
plt.ylabel("Mean PM2.5 (µg/m³)")
plt.xlabel("City")
plt.grid(True)
plt.show()
```

`<ipython-input-69-2a317c952b81>:3: FutureWarning:`
Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'x' variable to 'hue' and set 'legend=False' for the same effect.

`sns.barplot(data=pm25_only, x="city", y="mean", palette="Set2")`

City	Mean PM2.5 (µg/m³)
NYC	7.5
Boston	2.0
Chicago	12.5

```
[ ] # STEP 5: Show summary stats table
print("\n Summary Statistics (PM2.5 only):")
display(pm25_only[["city", "mean", "std", "min", "max"]])
```

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Email ID: U24102220@sharjah.ac.ae

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```
Summary Statistics (PM2.5 only):
```

	city	mean	std	min	max
0	NYC	6.896571	13.985542	0.000000	384.130000
2	Boston	1.837699	0.765489	0.464346	8.028841
4	Chicago	12.748455	12.057844	0.000000	2010.439941

```
[ ] print("Full Summary Table (All Pollutants):")
display(all_summary[["index", "city", "mean", "std", "min", "max"]])
```

```
Full Summary Table (All Pollutants):
```

	index	city	mean	std	min	max
0	pm25	NYC	6.896571	13.985542	0.000000	384.130000
1	pm10	NYC	18.069142	45.071864	0.000000	499.850000
2	pm25	Boston	1.837699	0.765489	0.464346	8.028841
3	no2	Boston	12.337025	1.503303	9.949419	15.802171
4	pm25	Chicago	12.748455	12.057844	0.000000	2010.439941

```
[ ] # Final Comparison Clustering + Regression

# Load aggregated H3-level data
nyc_h3 = pd.read_csv("nyc_aggregated_h3.csv")
boston_h3 = pd.read_csv("boston_aggregated_h3.csv")
chicago_h3 = pd.read_csv("chicago_aggregated_h3.csv")

# Combine all cities
all_h3 = pd.concat([nyc_h3, boston_h3, chicago_h3], ignore_index=True)

# Drop rows with missing data
features = ["pm25_mean", "temp_mean", "humidity_mean"]
X = all_h3[features].dropna().reset_index(drop=True)
X_scaled = StandardScaler().fit_transform(X)

# KMeans Clustering Across Cities
kmeans_all = KMeans(n_clusters=3, random_state=42)
all_h3_filtered = all_h3.dropna(subset=features).reset_index(drop=True)
all_h3_filtered["cluster"] = kmeans_all.fit_predict(X_scaled)

# Plot Clustering Across Cities
plt.figure(figsize=(10, 5))
sns.scatterplot(data=all_h3_filtered, x="pm25_mean", y="temp_mean", hue="cluster", style="city", palette="Set2")
plt.title("Cross-City Clustering: PM2.5 vs Temperature")
plt.xlabel("Mean PM2.5")
plt.ylabel("Mean Temperature")
plt.grid(True)
plt.show()
```

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